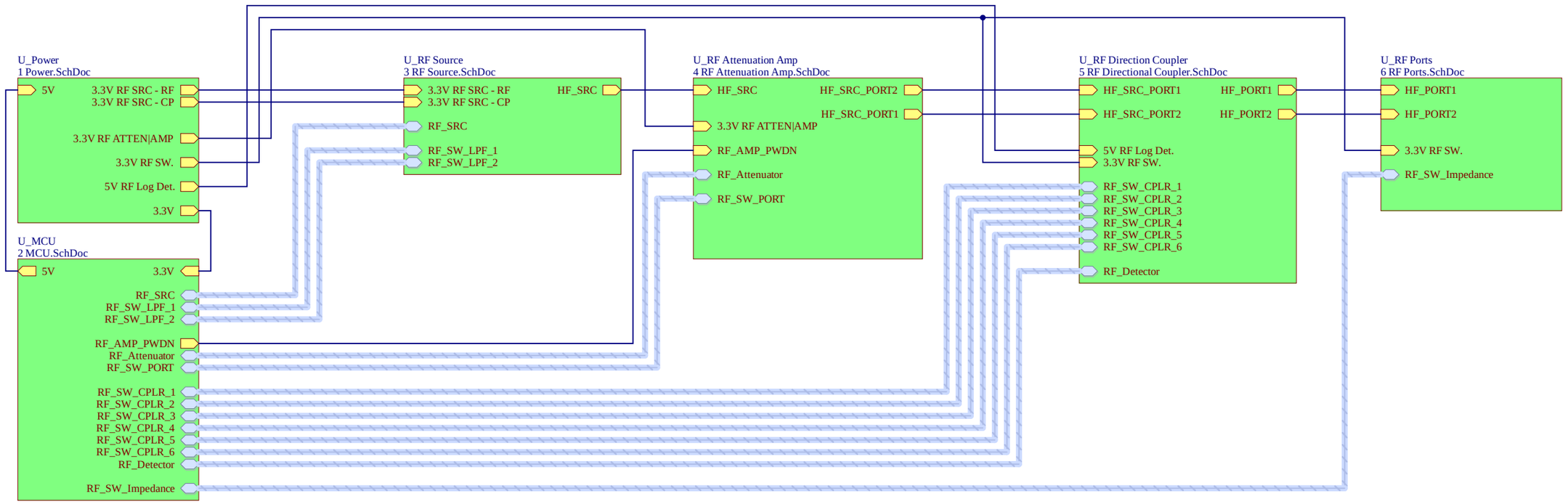


Top

FB7 Fiducial ° Fiducial Hadimec

FB3 STT SN: SN: GND

FB6 Fiducial ° Fiducial Hadimec

FB2 STT Logo Logo STT NG Tiny

FB9 Fiducial ° Fiducial Hadimec

FB1 STT Num STTNumber

Bottom

FB5 Fiducial ° Fiducial Hadimec

FB4 Fiducial ° Fiducial Hadimec

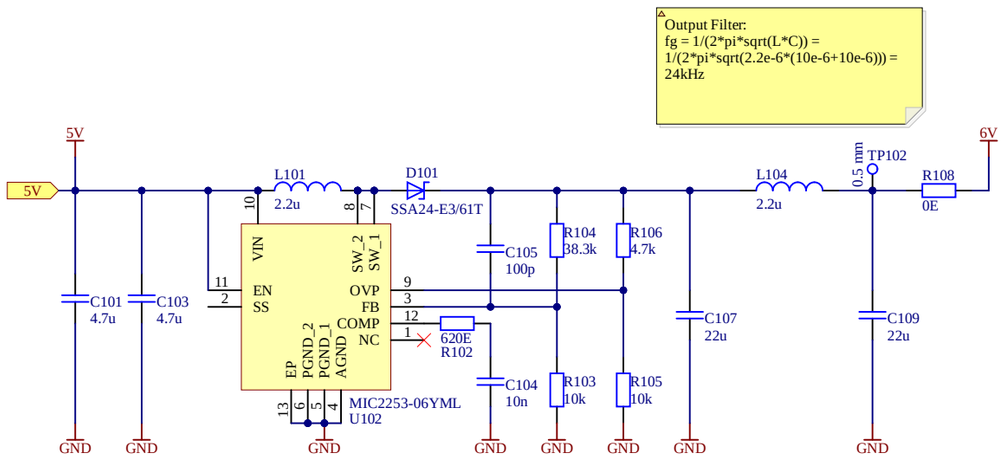
FB8 Fiducial ° Fiducial Hadimec

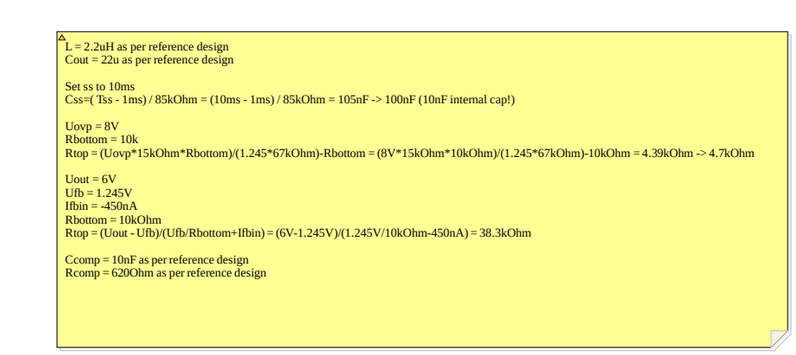
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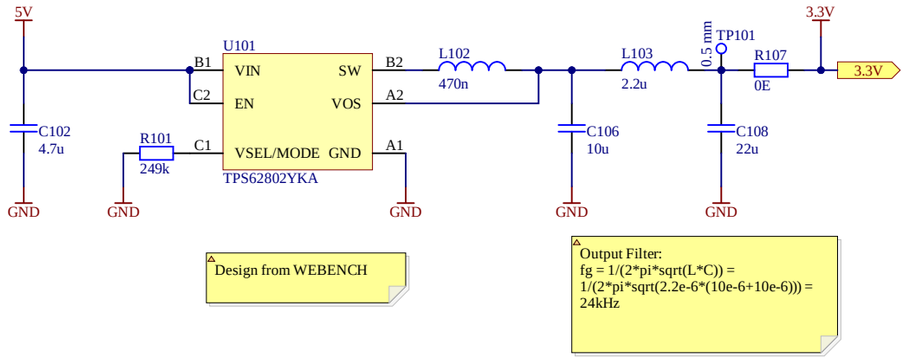
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Gibb, Bern

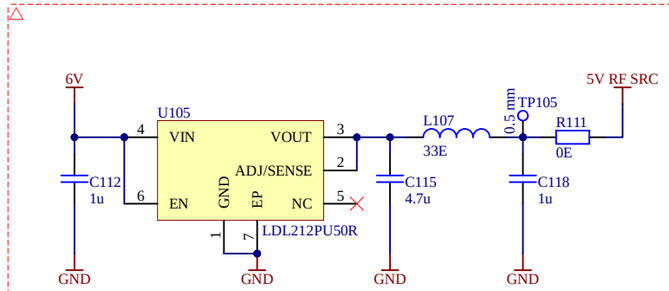
Project:	**SNA**	**Number:**	**SNA**	**Git:**	**4781f7d2 [Locally M**
Board:	**SNA**	**Variant:**	**EMS**	**Version:**	**1.0**
Page:	**Overview**	**Release:**	**dd.mm.yyyy**	**Sheet**	**1 of 7**
				Size:	**A3**
				Design:	**ML**
				Check:	**ML**

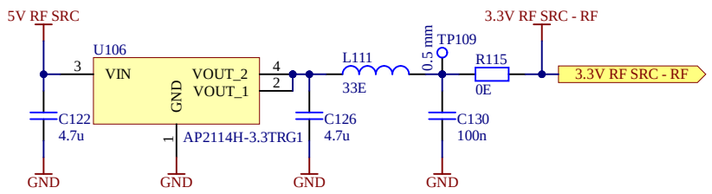
File: https://github.com/ml3rc/SNA.git 0 Overview.SchDoc

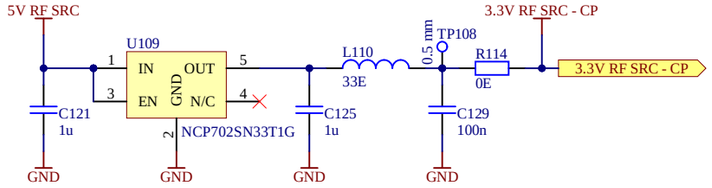


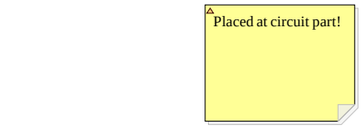


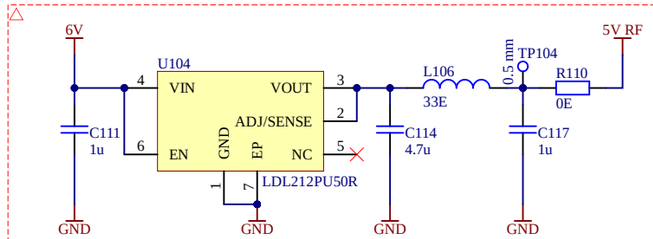


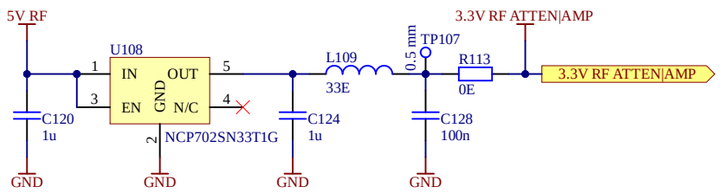


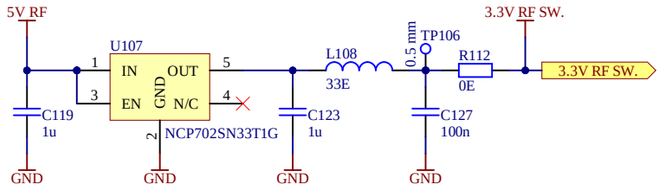




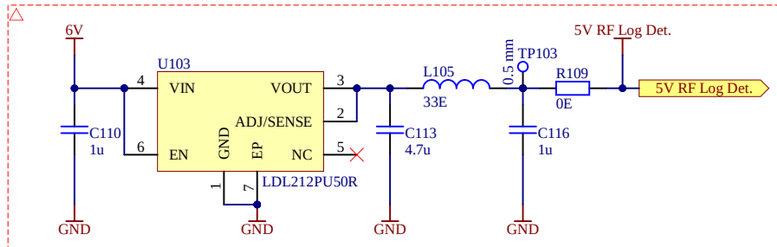


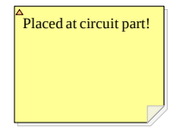


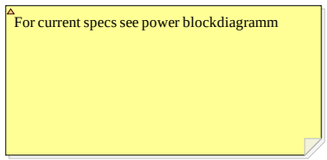








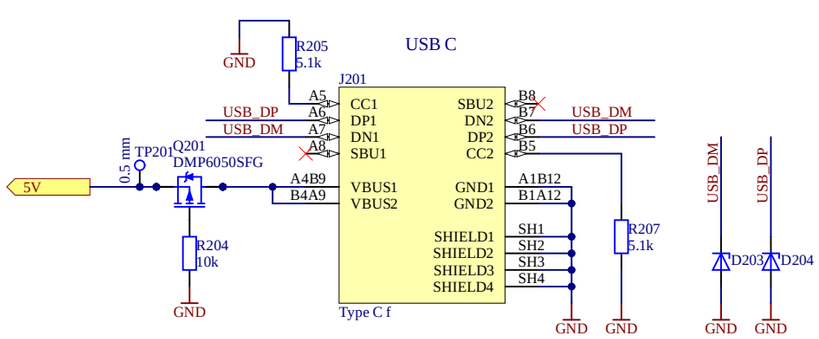


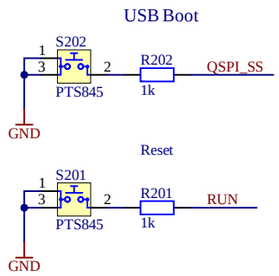


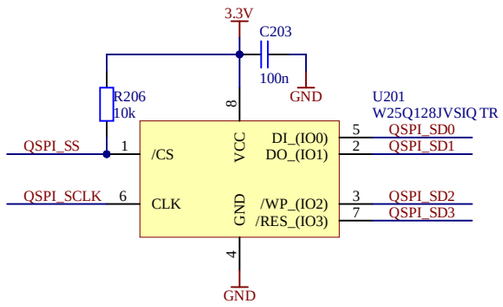
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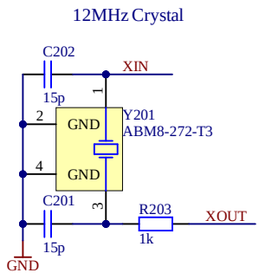
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** Power	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 2 of 7 **Size:** A3	**Git:** 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

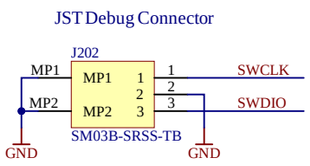
File: https://github.com/ml3rc/SNA.git 1 Power.SchDoc

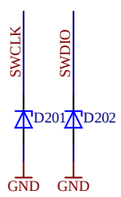


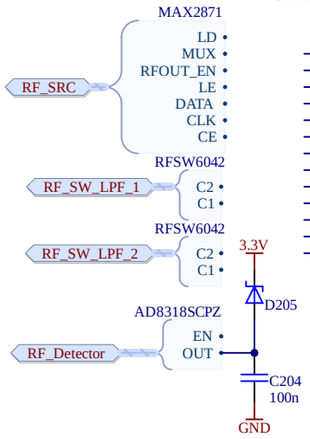


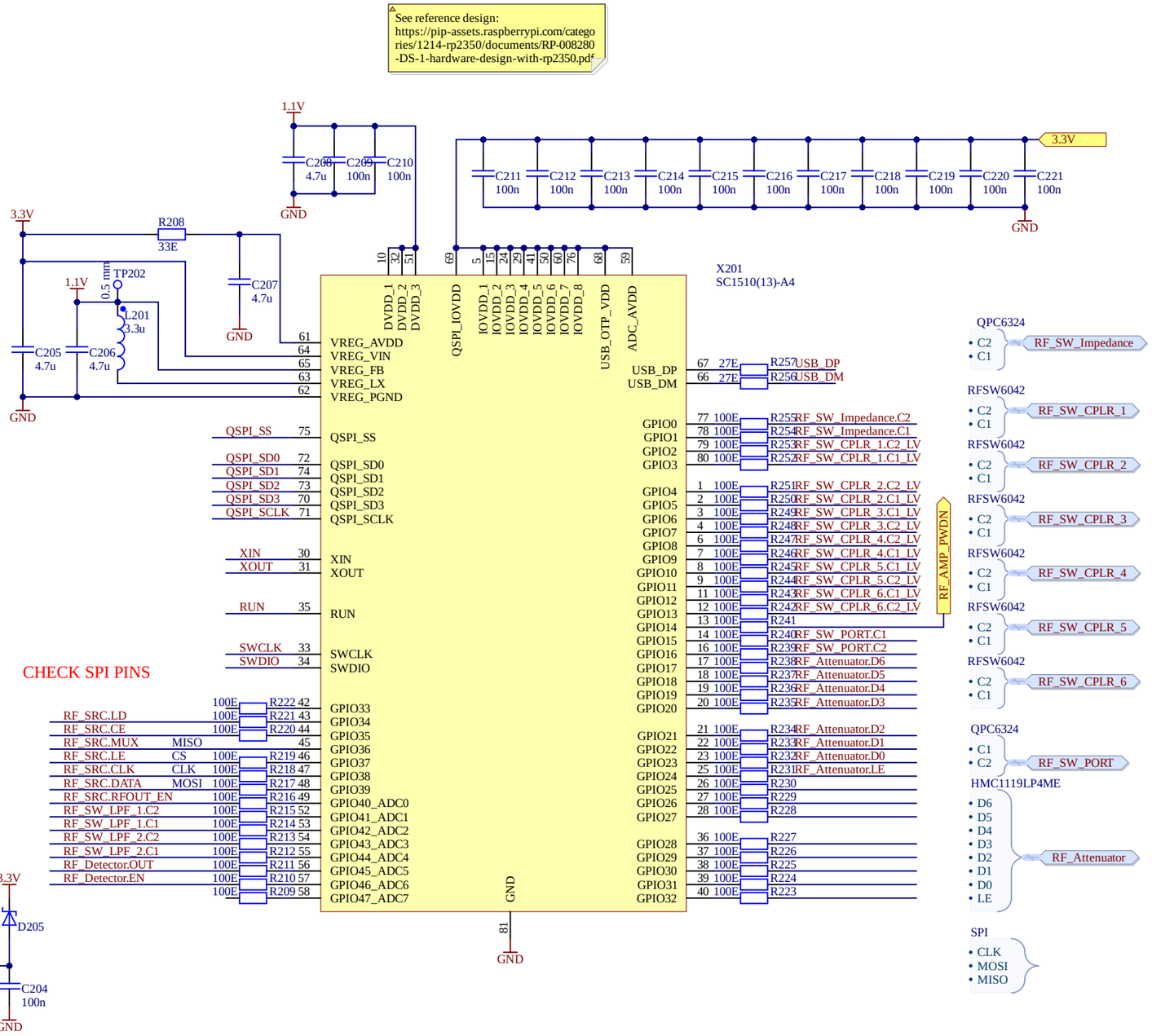










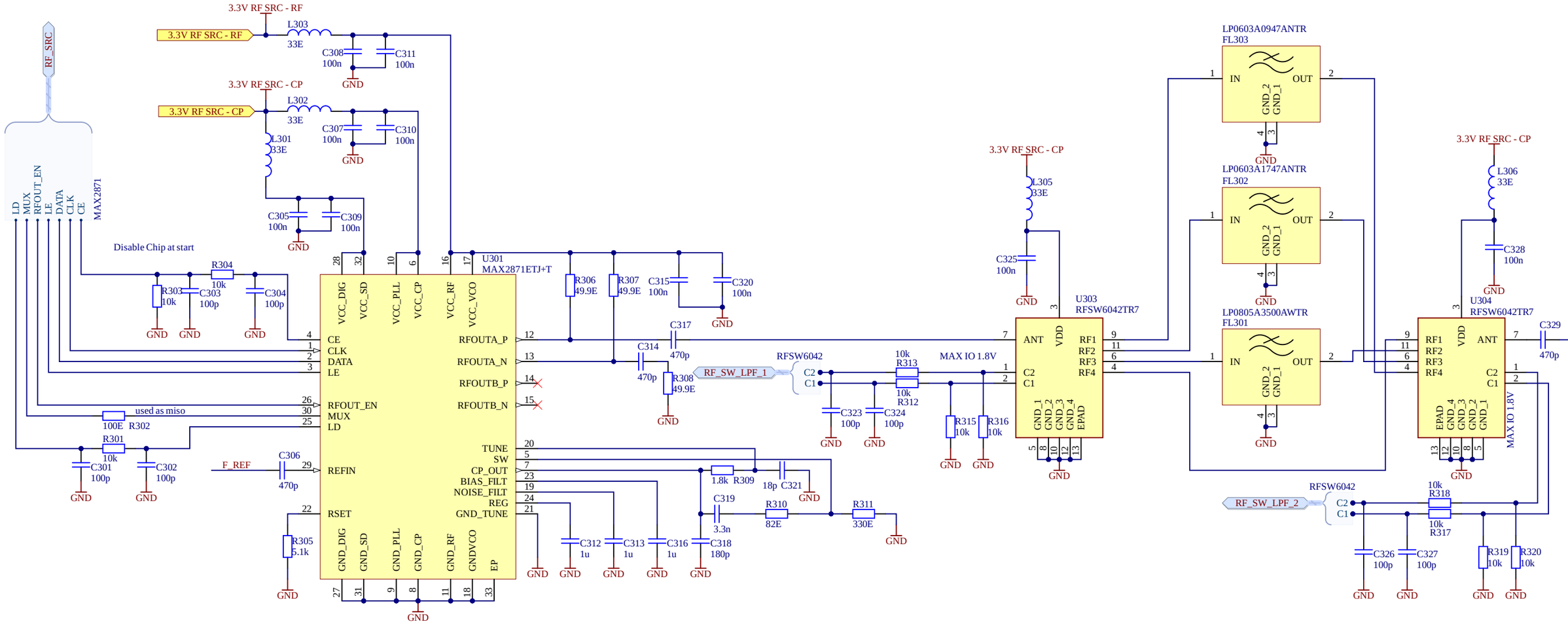


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Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** MCU	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 3 of 7 **Size:** A3
			Git: 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

File: https://github.com/ml3rc/SNA.git 2 MCU.SchDoc

See LibreVNA Schematics for reference



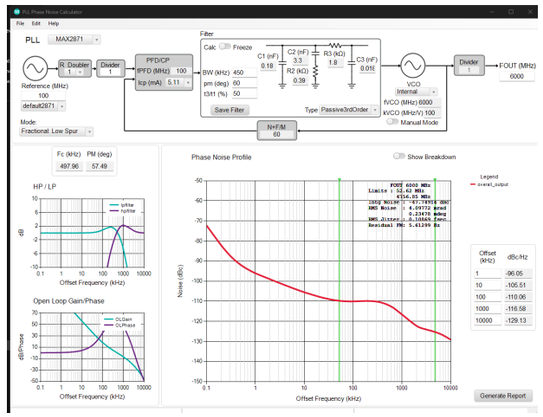
Rset = 5.1kOhm as per datasheet

C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

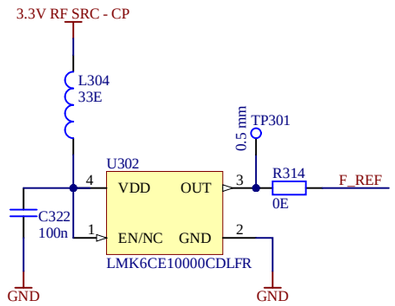
The loop filter can be calculated with:
https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

R_Crf_coupling = ((1)/(2*π*50MHz*470pF)) = 6.8Ohm



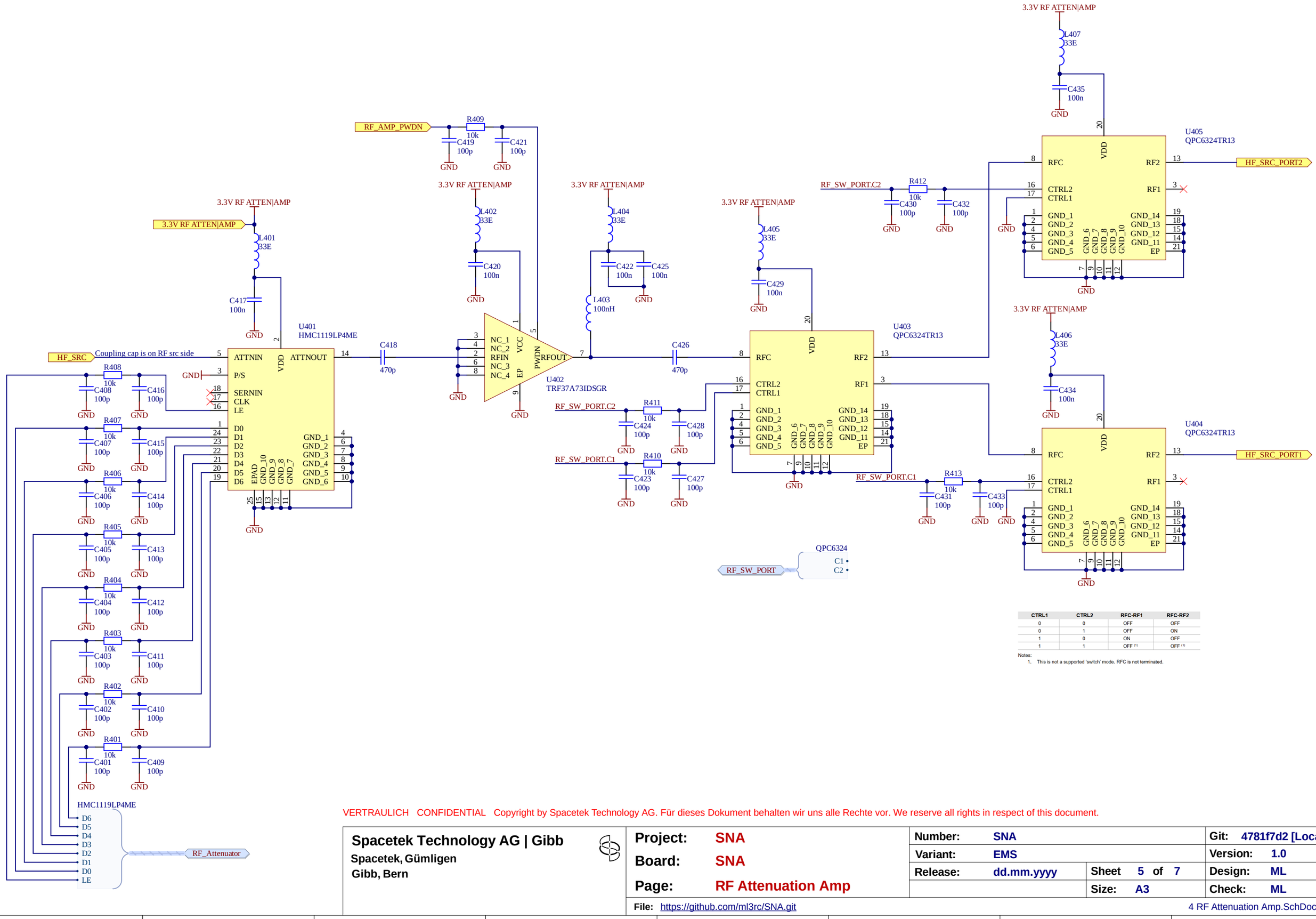
100MHz REF Clk.



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Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** RF Source	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 4 of 7 **Size:** A3	**Git:** 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

File: https://github.com/ml3rc/SNA.git 3 RF Source.SchDoc



CTRL1	CTRL2	RFC-RF1	RFC-RF2
0	0	OFF	OFF
0	1	OFF	ON
1	0	ON	OFF
1	1	OFF (1)	OFF (1)

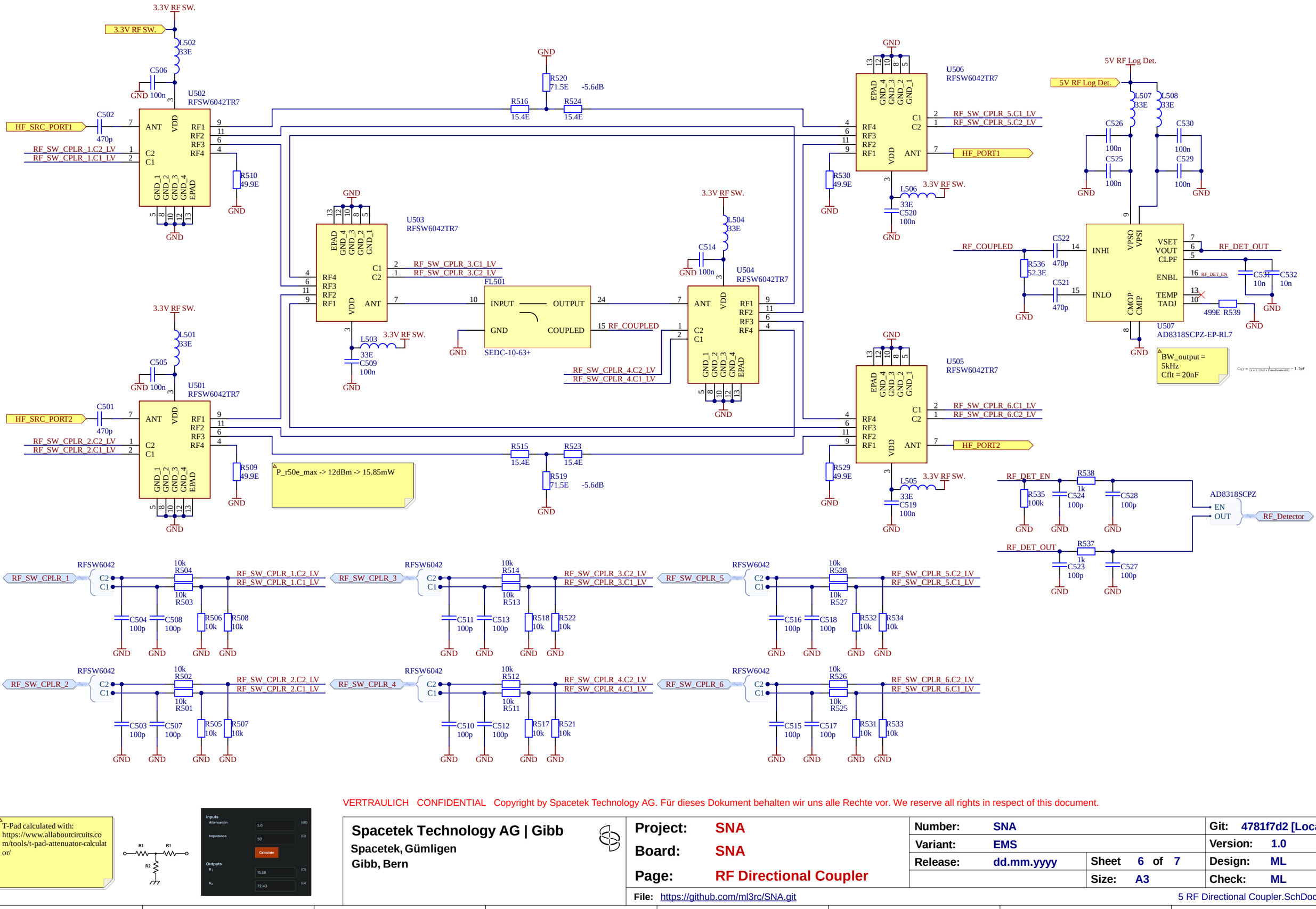
Notes:
1. This is not a supported 'switch' mode. RFC is not terminated.

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Project:	SNA	**Number:**	SNA	**Git:** 4781f7d2 [Locally M
Board:	SNA	**Variant:**	EMS	**Version:** 1.0
Page:	RF Attenuation Amp	**Release:**	dd.mm.yyyy	**Sheet** 5 of 7
				Size: A3
				Design: ML
				Check: ML

File: https://github.com/ml3rc/SNA.git 4 RF Attenuation Amp.SchDoc



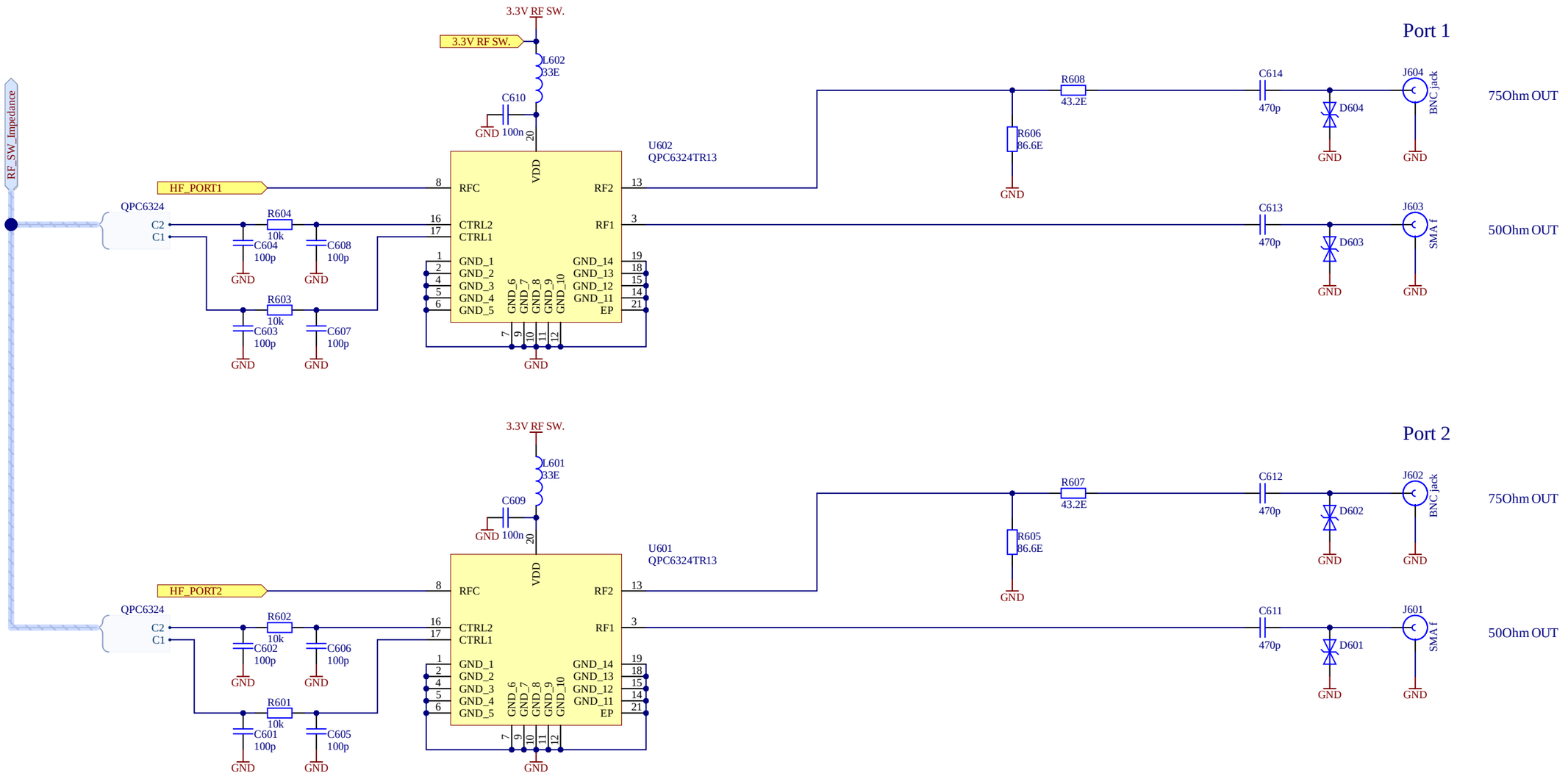
T-Pad calculated with: https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/

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Project:	**SNA**	**Number:**	**SNA**	**Git:**	**4781f7d2 [Locally M**
Board:	**SNA**	**Variant:**	**EMS**	**Version:**	**1.0**
Page:	**RF Directional Coupler**	**Release:**	**dd.mm.yyyy**	**Sheet**	**6 of 7**
				Size:	**A3**
				Design:	**ML**
				Check:	**ML**

File: https://github.com/ml3rc/SNA.git 5 RF Directional Coupler.SchDoc



50-75 Matching Pad:
solve({75=((1)/(((1)/(Rp))+((1)/(50))))+Rs,50=((1)/(((1)/(75+Rs))+((1)/(Rp)))))
Rp = 86.6Ohm
Rs = 43.3Ohm
Insertion Loss = 5.72dB

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Project:	SNA	**Number:**	SNA	**Git:**	4781f7d2 [Locally M
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	RF Ports	**Release:**	dd.mm.yyyy	**Sheet**	7 of 7
				Design:	ML
				Size:	A3
				Check:	ML

File: https://github.com/ml3rc/SNA.git 6 RF Ports.SchDoc